

Project Overview

Aims of the Project:

- The aim of the MCTR1021 course project is to generate optimal or near-optimal solutions for real-life problems related to **multi-agent cooperative systems** via meta-heuristic optimization techniques.
- Students can choose projects under the umbrella of two different themes:
 1. **Multi-agent cooperative systems in robotics and/or autonomous systems** (e.g. autonomous vehicles or UAVs, ... etc)
 2. **Renewable energy systems** (e.g. PV panels, wind farms, microgrids, ... etc).

Hardware:

- No hardware is required in this project context.

Programming Languages Allowed:

- You can choose either **MATLAB or Python** to implement your projects.
- All techniques, visualizations and simulations will be implemented using the chosen language.
- The chosen language **will not be changed** after the implementation of the first algorithm.

Teams:

The team formation regulations are:

- Each team will be composed of **4-5 members**.
- No exceptions will be made to form a team less than or more than the specified number of team members mentioned above.
- Students not assigned to teams by the deadline of teams registration will be randomly clustered or assigned into groups and added to the list of teams.
- No change to the project teams (interchange to another team or exchange of project) will be accepted under any circumstances.

Project Selection:

- Each team will be asked to review **5 recent scientific articles** (i.e. publication date between 2019-2021) related to the previously mentioned themes.
- From the reviewed papers, each team will select **2 different papers** from which they will submit project proposals.
- The proposal should be submitted in the **template** provided on CMS.
- The proposal should be **maximum 2 pages**.
- The course team will review both proposed ideas and one of them will be selected and allocated to the team.

Submission Deadline:

- The deadline of proposals submission is on **Tuesday, 19 October 2021, at 23:59.**

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Project Milestones:

- The overall maximum grade for the project is **40% from the course grade**.
- The project is split into **6 milestones** to facilitate and guide you in the implementation.

In the following table, the weight of each of the milestone will be presented.

Milestone	Brief Description	Deadline
1: Proposal (3%)	Two-page proposal of 2 project ideas within the scope of the previously mentioned themes.	Week 3: Tuesday, 19 October 2021
2: Formulation (5%)	Two-page literature survey of minimum 10 relevant research articles in the 5 preceding years (2017 – 2021) and Problem Formulation of the assigned project.	Week 5: Tuesday, 2 November 2021
3: SA (6%)	Implementation of Simulated Annealing (SA) algorithm on your problem.	Week 6: Thursday, 11 November 2021
4: GA (8%)	Implementation of Genetic Algorithm (GA) on your problem.	Week 9: Thursday, 2 December 2021
5: PSO / ACO/ WOA (8%)	Implementation of either Particle Swarm Optimization (PSO) or Ant Colony Optimization (ACO) or Whale Optimization Algorithm (WOA) on your problem.	Week 11: Thursday, 16 December 2021
6: New Algorithm + Final Report (10%)	Implementation of a new meta-heuristic population-based optimization algorithm (not taught in the course) on your problem. Final Project Report (6-page IEEE paper format – LaTeX). Optional Bonus Task: Implementation of machine learning techniques on your problem.	Week 13: Thursday, 30 December 2021

*The weight of each milestone is presented between brackets beside the name of each milestone.

Each milestone will be graded based on several criteria including **but not limited to**:

- Quality of all submitted material.
- For codes implementation: scalability, modularity, documentation (i.e. well commented), etc.
- For results analysis and discussion: analysis of all results (numerical, graphical, performance metrics, parametric studies, statistical studies, etc.) in light of the specific problem on-hand; as well as commenting on noticed trends and ability to highlight recommendations resulting from the analysis and discussion of results.
- For reporting: clarity of presentation and professional discussion with multiple aids (e.g. block diagrams, flowcharts, figures, tables, graphs, charts, etc.).